

AI-Enhanced Annotation Tool for Aneurysm Medical Image Labeling

Martin Paralič, Patrik Kamencay, Róbert Hudec

Department of Multimedia and Information-Communication Technologies

University of Zilina

Zilina, Slovakia

martin.paralic@uniza.sk, patrik.kamencay@uniza.sk, robert.hudec@uniza.sk

Abstract—Medical image annotation is crucial in developing artificial intelligence (AI) models to detect and analyze aneurysms. Traditional manual annotation methods are time-consuming, labor-intensive, and prone to variability between experts. To address these challenges, we propose an AI-powered annotation tool designed to mark and label aneurysms on medical images. This tool integrates deep learning techniques to assist radiologists and medical professionals in efficiently identifying and annotating aneurysms with high accuracy and consistency. It uses a model based on You Only Look Once (YOLO) version 11 for real-time aneurysm detection.

Index Terms—Aneurysm detection, cerebral aneurysm, annotation tool, artificial intelligence.

I. INTRODUCTION

The diagnosis and analysis of cerebral aneurysms represent a significant challenge in neuroradiology. Accurate detection of these pathological aneurysms in the blood vessels of the brain is crucial for the prevention of life-threatening complications such as subarachnoid haemorrhage [1].

Computed Tomography Angiography (CTA) or Magnetic Resonance Imaging (MRI) methods are predominantly used for diagnosis [2]. The examination uses high-powered X-rays in combination with several detectors, and the results are evaluated by computer. CTA and MRI scans are essential for accurate diagnosis and effective treatment of various neurological diseases and conditions. In the case of CTA, the area of the body being examined is exposed to a beam of X-rays [3]. The radiation passes through the tissue some of the radiation is absorbed, while the rest continues through and is detected by the scanner. The X-ray detector recognizes the different density levels of the tissue being scanned and transmits the data to a computer. The computer processes several consecutive X-ray images from various angles around the head. It combines them to produce cross-sections of the scanned veins, blood vessels, and soft tissues, so a CT scan provides more comprehensive information than a plain X-ray [4], [5]. Radiologists observe blood vessels on the monitor, displayed in varying shades of gray, with CTA scans generating either 2D or 3D images of

them [6]. A contrast dye is sometimes used to give a clearer view of the structures under examination.

Magnetic Resonance Imaging (MRI) is a diagnostic technique that uses a powerful magnetic field and high-frequency electromagnetic waves to create detailed images of the internal structures of the body [7]. During examination, the patient is placed within a powerful magnetic field and a brief radio-frequency pulse is applied to the body. This interaction generates electrical signals, which are captured and processed by a computer to produce a detailed image. MRI allows for the capture of cross-sectional slices of a specific area, which can be further processed or combined to create a three-dimensional (3D) representation of the blood vessels. The differentiation of various tissues in the images is based on their distinct responses to the same external conditions. MRI scans can be performed with or without a contrast dye, depending on the diagnostic requirements [8].

The acquired images are reviewed by a specialized radiologist, a doctor trained in diagnostic imaging techniques such as X-ray, CT, MRI, ultrasound, and nuclear medicine. The radiologist's primary role is to interpret medical images and provide diagnostic insights to other healthcare professionals. They typically use advanced software, such as PACS (Picture Archiving and Communication System), which facilitates detailed image analysis. This process is repeated regularly, as radiologists compare current images with previous scans to assess the progression or regression of any pathological changes. It is a meticulous, time-consuming task that requires attention to small details often hardly visible to the naked eye.

In recent years, Artificial Intelligence (AI) has become an essential component of medical imaging and image analysis. Various machine learning and deep learning models are employed for aneurysm detection, with convolutional neural networks (CNNs) and hybrid architectures being the most used. Well-known models, such as U-Net and its variations (including Attention U-Net and 3D U-Net), are widely used for aneurysm segmentation in MRI and CTA images [9]. Another commonly used model is ResNet (Residual Network), while hybrid CNN + RNN/LSTM architectures have also shown promise [10].

The promising approach is the application of deep learning models based on YOLO (You Only Look Once). In this paper, we present an annotation tool leveraging the YOLO v11

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model [11]. It assists with the detection of brain aneurysms in medical images. This system enables experts to efficiently label and validate findings, enhancing diagnostic accuracy and accelerating the analysis process. We discuss the model's architecture, its training process using medical datasets, and the practical benefits of its implementation in clinical practice. The annotated data can then be used to train better models on a large dataset of patients.

II. MATERIALS AND METHODS

Recent advancements in artificial intelligence (AI) have facilitated the automation of aneurysm detection. In this paper, a deep learning-based annotation tool leveraging the YOLO v11 model is employed to assist radiologists in the identification and annotation of cerebral aneurysms in CTA scans.

A. Data Collection

We are utilizing a small, anonymized CTA database provided by our research partner at the Department of Radiology, University Hospital in Martin, Slovak Republic. The dataset consists of angiographic scans from 11 patients diagnosed with various cerebral aneurysms. Each patient's records are stored in the Digital Imaging and Communications in Medicine (DICOM) format, and each one represents a record of a single patient. Each one contains 122 head angiographs captured from different rotational angles. The dataset images have a resolution of 1024×1024 pixels, and each pixel has a grayscale resolution of 10 bits. Informed consent was obtained from all study participants, allowing the use of their anonymized data for research purposes. The dataset includes brain aneurysm annotations provided by radiologist specialists, which are essential for training and validating machine learning models in medical image analysis. Aneurysms were highlighted as rectangular regions of interest. The majority of patients had a single aneurysm, while one case presented with two aneurysms, and another case involved multiple aneurysms. Since CTA scans capture aneurysms from multiple angles, we have several views of the aneurysm. This diversity in imaging helps the neural network learn from a broader range of data, improving its ability to detect and analyze aneurysms effectively.

Prior to model training, the images underwent several pre-processing steps:

- **Normalization:** Pixel intensity values were normalized to ensure uniformity across the dataset.
- **Resizing:** Images were resized to a standard dimension suitable for input into the YOLOv11 model.
- **Augmentation:** Techniques such as rotation, scaling, and flipping were applied to augment the dataset, enhancing the model's robustness to variations in image orientation and size.

B. YOLO v11 model

The YOLO v11 model, a convolutional neural network (CNN)-based object detection framework, is implemented for real-time aneurysm detection. The model is trained using a

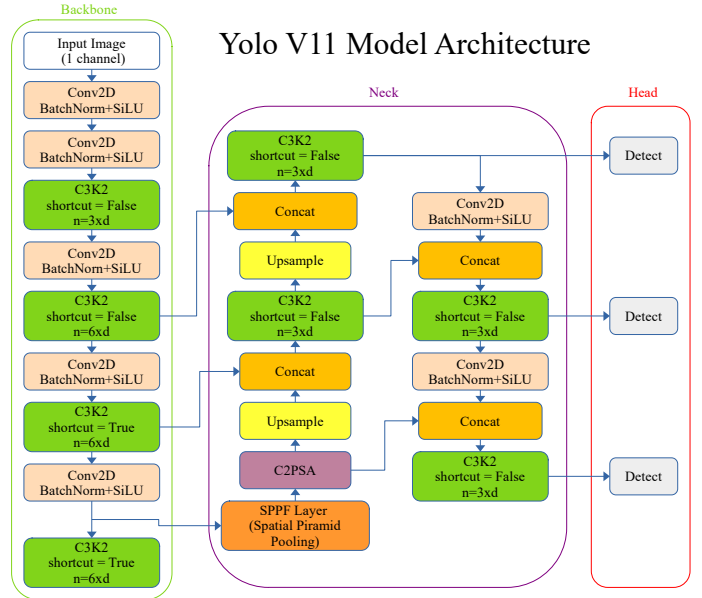


Fig. 1. Architecture of model Yolo V11.

curated dataset of annotated medical images, ensuring robust feature extraction and accurate classification of aneurysmal structures. The training pipeline includes:

- **Data Preprocessing:** Standardization of image resolution, normalization, and augmentation techniques such as rotation, flipping, and intensity adjustments.
- **Annotation Process:** Radiologists specialist manually annotate aneurysms using bounding boxes, creating a high-quality labeled dataset for supervised learning.
- **Model Training:** The YOLO v11 network is trained using back-propagation and Stochastic Gradient Descent (SGD), optimizing parameters to enhance detection accuracy.
- **Evaluation Metrics:** The performance of the model is assessed based on precision, recall, F1-score, and mean Average Precision (mAP) to ensure reliability in clinical applications.

The Fig. 1 illustrates the architecture of the model used. In object detection models, a backbone network plays a crucial role in extracting and encoding features from the input data. It serves as a fundamental feature extractor, capturing both low-level and high-level features. Additionally, a neck component further processes and refines these extracted features, enhancing their representation to provide more informative outputs.

C. Model Preparation

Before training the model, we randomly split the data into three sets: training, testing, and validation. Each image was only in one set at a time. The division ratio was as follows:

- **Training** – 60% images / 854 files.
- **Testing** – 20% images / 244 files.
- **Validation** – 20% images / 244 files.

1	0	0.456055	0.502930	0.029297	0.025391
2	0	0.442871	0.517578	0.014648	0.011719
3	0	0.435547	0.502441	0.023438	0.026367
4	0	0.429688	0.533203	0.027344	0.025391
5					

Fig. 2. Example of Yolo annotation, where patient has multiple aneurysms.

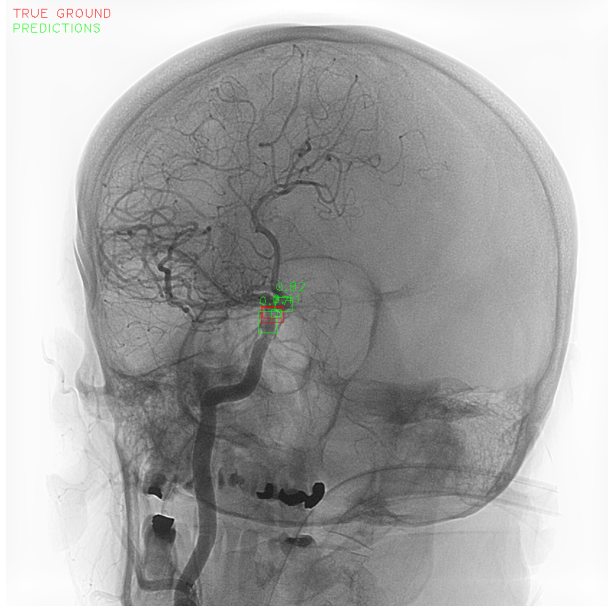


Fig. 3. Trueground annotation vs. aneurysm detector.

To train the model, we extracted images from DICOM format files and adapted the annotation format to Yolo. The format is relatively simple, consisting of a class number followed by the relative position of the center, as well as the relative height and width of the region of interest. In the case of multiple annotations, additional records with the same structure appear on subsequent lines, as show Fig. 2. The class configuration information is stored in a separate .yaml configuration file.

These manual annotations are software plotted into the original medical image as "TRUE GROUND" annotations and are indicated by red rectangles, as shown in Fig. 3. The figure also includes the automatic detections generated by the Yolo detector, marked as green rectangles "PREDICTIONS". The number above the rectangle represents the value of the detection reliability. The model was trained using a supervised learning approach:

- Training Set: A portion of the dataset was designated for training, with images annotated by experienced radiologists to provide ground truth labels.
- Validation Set: A separate subset was used for validation to monitor the model's performance and prevent overfitting.

Table I numerically expresses the results of model detection

TABLE I
EVALUATION RESULTS AND COMPARISON WITH OTHER STUDIES.

AI Model	Metrics		
	Precision	Recall	F1-Score
Yolo v11n ^a	0.979	0.939	0.958
Yolo v8n ^a	0.924	0.909	0.917
1D-CNN [12]	0.952	0.952	0.957
3D-CNN [13]	0.958	0.943	0.951
CNN/MIP [14]	0.943	0.937	0.939

^aAuthor results.

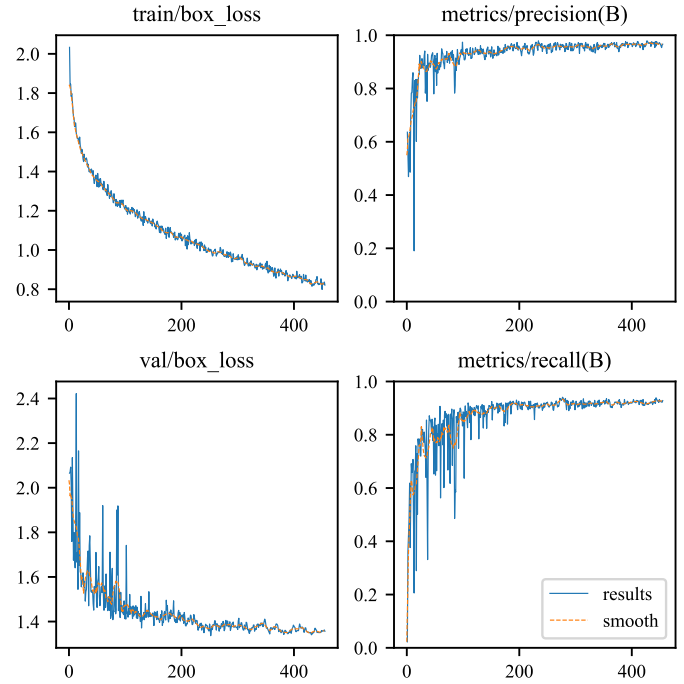


Fig. 4. Training and validation results.

testing against manually annotated data. The first row represents the evaluation of the proposed model. The second row shows our initial results when we worked with the Yolo v8 model. The following rows compare the results of different models from several authors performing a similar study. The results are comparable, but the main advantage of the Yolo models is speed, where classification for the whole image is available in less than 300ms. The flow of training and testing shows Fig. 4. After about 200 epochs, precision and recall have already reached their maximum. The validation error stabilized a few epochs later.

III. IMPLEMENTATION

The proposed application is programmed in a Python environment. The graphical environment is created using the PyQt6 cross-platform library. The application has been tested under Windows 11 and Debian. It uses cross-platform libraries and should work on any version of Windows, Linux. We have not tested for Mac OS, but there are no prerequisites used that would prevent this application on Mac OS platform. To run application use virtual environment with requirements.txt.



Fig. 5. Application design, implementing Yolo V11 model.

The original data are not publicly available. Only anonymized test samples can be found in the project. The proposed application with the tested models is publicly available on GitHub: <https://github.com/martinparalic/An2025>. Fig. 5 shows the graphical user interface. On the left is a floating toolbar, which offers basic functions such as working with files and views. There is also functionality for auto-detection of the aneurysm, respectively the user can add a manual annotation. In the middle of the screen is the medicine image itself. The primary resources were DICOM files. However, standard image formats like JPG, PNG, BMP are supported. Format compatibility is consistent with the supported formats of the pydicom and opencv-python libraries.

The panel on the right offers functions to display a multi-bit grayscale image, into false color mode. Since the standard output (screen) only displays the grayscale images as an 8-bit grayscale image, we can map the remaining bits into false colors. In false colors, paint details can be highlighted. Followed by a combo box for model selection. In the paper, we presented only the best model based on Yolo V11. The application has several models to choose from, which were created during the search for optimal parameters, including older versions of the Yolo V8, which we do not present in this article. You can add another model to the application in a simple way - by copying the new pytorch model to the "models" folder. The application automatically detects existing models in this folder when it starts. This is followed by a combo with a classes and a list of the annotations themselves.

IV. CONCLUSION

In this paper, we developed an AI-powered annotation tool designed to assist radiologists in identifying and labeling aneurysms in medical images. By integrating deep learning techniques, specifically the YOLOv11 model, the tool en-

hances the efficiency and accuracy of aneurysm detection. Our results demonstrate that AI-assisted annotation significantly reduces the time required for labeling while maintaining high consistency and accuracy. The developed tool has the potential to streamline the annotation process, thereby contributing to the advancement of AI models for aneurysm detection and analysis.

To further enhance the capabilities of the AI-powered annotation tool, we propose the following future directions:

- **Collect More Data:** Expanding the dataset by incorporating a larger and more diverse collection of medical images will improve model robustness and generalizability. This includes obtaining images from different institutions, patient populations, and imaging protocols.
- **Retrain Models:** Regularly updating and retraining the YOLOv11 model with newly annotated data will help refine its detection capabilities and reduce biases. Implementing transfer learning techniques may also allow for improved performance with limited additional data.
- **Expand to More Modalities:** Extending the tool's functionality beyond CTA images to other imaging modalities, such as digital subtraction angiography (DSA), will make it more versatile and applicable across a broader range of clinical scenarios.
- **Enhance User Interface and Workflow Integration:** Further development of the tool's interface to optimize usability, including better annotation review mechanisms and seamless integration with existing radiology workflows, will improve adoption among medical professionals.

The reliability of deep neural networks in aneurysm detection depends on several factors, such as the quality of the training data, the type of imaging methods (CT, MRI, angiography), and the model architecture. Research suggests that DNNs can achieve high accuracy, often comparable to experienced radiologists. However, challenges still exist. Advantages of DNNs in aneurysm detection:

- **High accuracy** - some studies report accuracy of over 90% in detecting cerebral aneurysms.
- **Automation** - Models can help in screening and reduce the workload of radiologists.
- **Capturing fine details** - DNNs can identify even small aneurysms that might be missed by a human.

Limitations and challenges:

- **False positive and negative findings** - Models can occasionally label normal structures as aneurysms or, conversely, miss true positives.
- **Dependence on data quality** - Models trained on limited or biased data may have lower reliability.
- **Explainability** - Deep neural networks often function as a "black box," making it difficult to understand the reasons behind their decisions.

Clinical Credibility:

- DNNs are used as a supportive tool for radiologists, not as a stand-alone diagnostic method.

- The combination of AI and humans leads to the best results, as DNNs can increase the accuracy and speed of analysis.
- DNNs are promising in aneurysm detection and can greatly help in clinical practice, but they do not yet replace the expert decisions of physicians.

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